

Optimal pyranometer placement in bifacial PV plants on complex terrain

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ABSTRACT

Accurate placement of irradiance sensors is critical for performance monitoring in utility-scale photovoltaic (PV) plants, particularly those featuring single-axis trackers, bifacial modules, and non-uniform terrain. Installing pyranometers on every tracker row is infeasible, making optimal sensor selection a key design challenge. In this work, we benchmark seven pyranometer placement algorithms—including geometric heuristics, unsupervised clustering, metaheuristics, and multi-objective optimization using a detailed simulation framework that incorporates clear-sky irradiance modeling, terrain-induced shading, backtracking dynamics, and bifacial rear-side contribution. The methods are evaluated across three operational PV plants in Spain, ranging from 30 to 70 MWp, each characterized by complex topography. We assess each algorithm using multiple performance metrics: mean absolute error (MAE), mean relative error (MRE), temporal correlation (R^2), and inter-sensor redundancy. Results show that multi-objective algorithms, particularly those incorporating simulated irradiance, consistently outperform geometry-only approaches in both accuracy and robustness. Notably, a geometry-based multi-objective method achieves comparable performance. We find that simple dispersion heuristics fail to generalize under steep terrain, while simulated annealing offers a strong trade-off between accuracy and runtime. These findings support a hybrid sensor placement strategy combining fast geometric pre-selection with energy-based refinement. The proposed methodology is scalable and applicable to modern PV systems, offering a reproducible framework for data-driven sensor deployment in heterogeneous landscapes.

1. Introduction and Motivation

As photovoltaic (PV) plants increase in size and complexity, accurate monitoring of solar irradiance becomes essential for evaluating operational performance and performance ratio (PR). Pyranometers are the standard instruments for measuring plane-of-array (POA) irradiance, which directly affects energy yield estimation, fault detection, and asset management strategies. In utility-scale facilities with hundreds or thousands of tracker rows, installing pyranometers in every row is neither economically nor logistically feasible. Therefore, selecting a representative subset of trackers for sensor placement becomes a critical task for plant designers and operators [1,2].

Traditional methods for selecting pyranometer locations are often based on simple geometric heuristics, such as choosing central rows or avoiding edge effects [3]. However, these approaches fail to account for the complex interaction between terrain, orientation, shading, and spatial variability in irradiance. With the widespread adoption of

bifacial modules and single-axis tracking systems, site-wide irradiance differences can become significant, particularly in undulating or mountainous terrain. Single-axis trackers require more sophisticated placement algorithms that incorporate site topography, dynamic shading analysis, and bifacial effects to accurately capture the irradiance distribution across the plant. Recent literature, including [4,5] highlight the need for advanced modeling and tailored sensor placement approaches to fully exploit the capabilities of bifacial modules on complex terrain

In parallel, machine-learning (ML) techniques have begun to inform sensor-placement strategies. Supervised models such as gradient-boosted trees have been trained on multi-station datasets to predict POA residuals and flag tracker rows that merit instrumentation [6]. In addition, graph neural networks (GNNs) are emerging as digital twins that reconstruct full-field irradiance from sparse inputs. By ranking node sensitivities, GNNs identify minimal yet information-rich sensor grids—a concept already validated in hydraulic and wind-energy networks

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and now under evaluation in PV applications [6,7]. These data-driven approaches complement the deterministic and metaheuristic algorithms benchmarked in this work and point toward hybrid workflows in which geometric pre-filters support adaptive, learning-based sensor placement.

The integration of complex terrain modeling into sensor placement strategies remains limited in the literature [8]. While some studies have addressed sensor allocation in rooftop or fixed-tilt systems on varied topography [4,9] few have examined the problem specifically for large-scale bifacial plants with single-axis trackers on mountainous terrain [10,11]. The challenge is compounded by the dynamic nature of tracking systems [10,12,13], where optimal sensor positions must account for continuously changing module orientations throughout the day.

Recent work by [14] provided comprehensive validation of View-Factor and Ray Tracing models for bifacial irradiance prediction, demonstrating the critical importance of accurate rear-side modeling. However, their focus was on model validation rather than sensor placement optimization. Similarly, [11], conducted extensive validation of bifacial PV simulation software but did not address the specific challenge of optimal sensor deployment.

This work addresses several critical gaps in the current literature:

1. **Comprehensive benchmarking** of geometric, heuristic, and multi-objective sensor selection algorithms specifically for utility-scale PV systems with single-axis trackers, bifacial modules, and complex terrain.
2. **Integration of terrain-induced shading** with dynamic tracking geometry and bifacial modeling into a unified simulation and evaluation framework.
3. **Validation against real operational data** from three utility-scale PV plants in hilly terrain, providing practical insights for industry implementation.
4. **Development of hybrid sensor placement strategies** that combine computational efficiency with energy-based optimization, suitable for large-scale deployment.

To the best of our knowledge, this is the first work to benchmark geometric, heuristic, and multi-objective sensor selection algorithms in utility-scale PV systems that include single-axis trackers, bifacial modules, and non-planar terrain. While previous studies have addressed sensor allocation in fixed-tilt or rooftop systems, this work integrates terrain-induced shading, dynamic tracking geometry, and bifacial modeling into a unified simulation and evaluation framework.

In this work, we show that algorithms based solely on geometric and topographic criteria can yield pyranometer selections that perform comparably to those obtained through energy-based optimization. To support this claim, we compare seven selection algorithms—ranging from unsupervised clustering to metaheuristic optimization—using both geometric/topographic metrics and simulated energy data. By incorporating shading, solar tracking, bifaciality, and terrain modeling, we present a reproducible and scalable framework for sensor placement in utility-scale PV plants.

2. Methodology

This study proposes and compares various methods for the optimal selection of pyranometer locations in utility-scale photovoltaic (PV) plants. The methodology relies on real design data and considers topographic, geographic, geometric, and energetic aspects. Seven selection algorithms, both heuristic and algorithmic, have been analyzed to evaluate their ability to represent the plant's average effective irradiance using a limited number of sensors. All the calculations and modelling were conducted via a Python script, in a computer Dell Precision 3480, that runs a Intel i7 processor.

2.1. Input data and Preprocessing

Geometric data was extracted from the Bill of Materials (BOM) exported in Excel format from commercial design software. For each tracker, the dataset included X and Y coordinates as well as the terrain elevation (Z) at each pole was obtained. Additional information on the location of combiner boxes (comboxes) was also included manually. The data were preprocessed to fill missing values, convert slopes from degrees to percent grade, and group rows by table identifier.

A digital elevation model (DEM) of the site in raster (tiff) format, derived from actual topographic surveys, was used to simulate shading effects using hourly hillshade maps, following the method described in [15].

2.2. Calculation of slopes and relevant distances

The north-south slope (tilt) of each table was calculated directly from the BOM data. The east-west slope was estimated as the height difference with the nearest neighbor in the east-west direction using the *NearestNeighbors* method from *sklearn.neighbors* library [16].

To quantify terrain complexity, slope histogram for three projects is provided in Section 2.5. This computation takes into account NS and EW slope of tables computed at its central point.

Additionally, the distance from each table to the closest combobox was computed as a proxy for connection feasibility. It was assumed that comboboxes are located at the medium-voltage stations given that all three projects present string inverters.

2.3. Irradiance modeling and shadow simulation

The PVLib library [17] was used to compute clear-sky irradiance components—global horizontal irradiance (GHI), direct normal irradiance (DNI), and diffuse horizontal irradiance (DHI)—for one representative day per month, at hourly resolution (288 h total). Solar position was determined using *pvlb.solarposition* method, and clear-sky irradiance was estimated using the Ineichen model [18] which offers a good trade-off between simplicity and physical accuracy under clear-sky conditions.

The limitation to clear-sky conditions is acknowledged as a simplification that may not fully represent real-world performance under cloudy or mixed-sky conditions. However, clear-sky modeling provides a consistent baseline for comparing sensor placement algorithms without the additional complexity introduced by cloud dynamics, which would require stochastic modeling approaches beyond the scope of this study. Front-side plane-of-array irradiance $G_{front}(t)$ was computed for each tracker using *pvlb.irradiance.get_total_irradiance*, which combines direct, diffuse, and ground-reflected components projected onto the tilted tracking surface. The input parameters include the tracker's hourly tilt and azimuth angles (derived from backtracking geometry), solar zenith and azimuth angles, and the site's albedo. Albedo has been considered constant and equal to 0.25 during the year.

To represent the impact of terrain-induced shading on $G_{front}(t)$, dynamic hillshade maps were generated every 3 h with pixels of 5 m using *gdaldem hillshade*. This tool calculates illumination conditions on a digital elevation model (DEM) by simulating the apparent solar angle, using azimuth and elevation values obtained from PVLib. The 3-hour temporal resolution represents a compromise between computational efficiency and accuracy, as terrain shadows change relatively slowly compared to tracking dynamics. Given limitations in the GDAL Python bindings when handling dynamic inputs, the command-line interface was used for better reliability [19].

Each tracker's shadow exposure was then estimated by interpolating its location on the hillshade raster. Shadow values were normalized to range from 0 (complete shadow) to 1 (full illumination). These discrete 3-hour values were linearly interpolated to match the 288-hour time series. An example of these shadow maps can be found in Fig. 1.

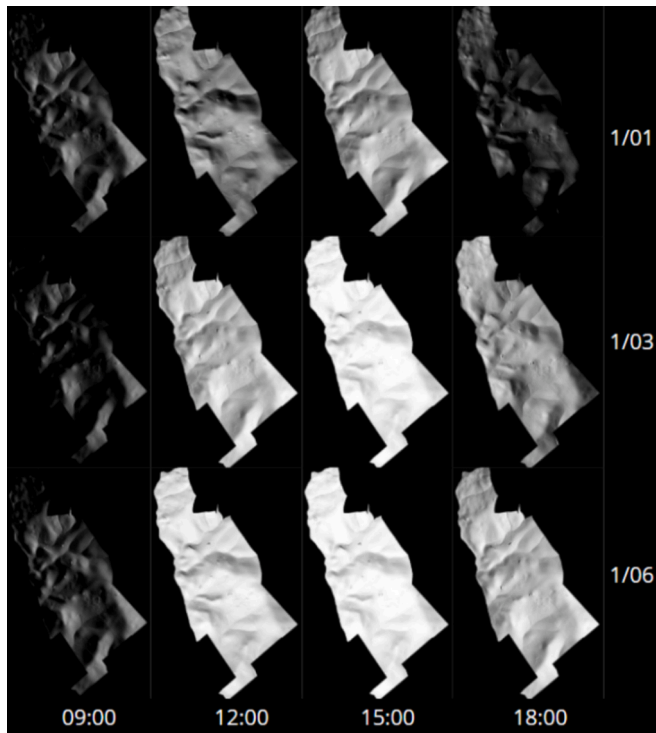


Fig. 1. Example of dynamic hillshade raster generation. In this example, 3 days of the year are simulated for different time of the day.

The resulting shadow factor $S(t)$ was applied as a multiplicative correction only to the direct irradiance component, as terrain shadows predominantly affect beam radiation. This is justified because terrain-induced shadows geometrically block only the line-of-sight between the sun and the module, suppressing direct or beam irradiance while leaving diffuse horizontal irradiance largely unaffected, except for extreme reductions in sky view factor [20]. The corrected front-side irradiance was thus computed as:

$$G_{front}(t) = G_{dir}(t) * S(t) + G_{diff}(t) * TVF + G_{refl}(t)$$

where G_{dir} , G_{diff} , and G_{refl} represent the direct, diffuse, and ground-reflected components, respectively, and TVF is the tracker view factor, computed from the DEM using the *Relief Visualization Toolbox (RVT)*. TVF corrects the diffuse component based on local terrain occlusion [21].

To model the bifacial rear-side irradiance $G_{rear}(t)$, the *pvfactors-timeseries* model was used [22]. This view-factor-based model computes irradiance exchanges between PV rows and the ground, accounting for tracker tilt, height, inter-row spacing (via ground coverage ratio) and surface reflectance (albedo) *geometry*. *pvfactors* inherently models inter-row self-shading, horizon visibility, and anisotropic reflections. This model assumes planar terrain and constant row geometry, which limits its ability to represent complex relief or non-uniform spacing between tables.

This simplification has several practical implications in real-world utility-scale PV plants.

- **Local tilt variations:** In sloped or undulating sites, the tilt and elevation of each PV row may differ substantially, leading to changes in both self-shading and the solid angle under which each row “sees” the ground and neighboring rows.
- **Ground distances and albedo:** The distance between module rear surface and ground, which impacts the proportion of reflected irradiance reaching the rear, can vary by > 20 % in highly non-planar terrain

- **Non-uniform mutual shading:** In reality, mutual shading between rows on rolling terrain is highly variable, as some zones will experience more/less obstruction of diffuse light or ground reflection than others [14]
- **Validation evidence:** Recent studies comparing *pvfactors* outputs to empirical measurements in real bifacial tracker plants on heterogeneous terrain report non-negligible systematic errors on the rear side (typically 5–15 %), especially in valley or ridge portions of a site.

Finally, the total irradiance per tracker was calculated as:

$$G_{tot}(t) = G_{front}(t) + G_{rear}(t)$$

This formulation integrates both front-side losses due to topographical shading and rear-side gains from bifaciality, providing an approximated representation of the irradiance environment under non-flat terrain. Note that no bifaciality factor is considered, since the actual irradiance has been estimated

2.4. Pyranometer selection algorithms

Selecting optimal pyranometer locations within a utility-scale PV plant requires careful consideration of multiple factors, including spatial layout and terrain variability. To systematically address this challenge, we implemented and benchmarked seven different selection algorithms. These algorithms span a wide range of complexity—from simple geometric heuristics to advanced metaheuristic optimizations—providing a comprehensive analysis of potential strategies for sensor placement. The following subsections describe each method in detail, highlighting their underlying principles, strengths, and limitations.

2.4.1. K-Means

The classic K-Means partitioner is applied to the five-dimensional space $\{X, Y, NS - slope, EO - slope, STS distance\}$. Because each cluster is forced to yield exactly one tracker, the method guarantees a wide spatial spread at linear cost $O(N k)$, a property that still motivates its use in very recent irradiance-forecast pipelines [23]. Its simplicity, however, comes at the price of assuming roughly hyper spherical clusters; when the actual topography is elongated, K-Means may over-represent flat zones while missing narrow valleys where shading gradients are highest.

2.4.2. Refined K-Means

The two-step variant first groups trackers by position and STS distance, then inside each cluster selects the tracker whose slopes and distance deviate least from the local mean. Such hierarchical refinements have been proven effective for bifacial façade optimization [24]), and here they confer better topographic representativeness than plain K-Means without inflating the computational budget; nevertheless, the ad-hoc scoring can overweight either slope or distance if the two variables are differently scaled, so pre-normalization is essential.

2.4.3. Hierarchical clustering

Ward-link agglomeration successively merges trackers so that the increase in total within-cluster variance is minimal. Ward’s metric naturally captures anisotropic terrain shapes and, as recently shown by [9] in a 320 MW plant, aligns well with sky-view-factor zoning for pyranometer siting. Yet the method scales quadratically with the number of trackers and can be hampered by the classical chaining effect, making it less practical for portfolios exceeding ~10 000 trackers.

2.4.4. Maximum-dispersion

Starting from a random tracker, the greedy farthest-point sampler iteratively adds the candidate that maximizes its minimum distance to the current subset, while penalizing slope and distance deviations. Farthest-point ideas remain competitive. The algorithm excels at geometric coverage and contains no hyper-parameters; however, because it

is blind to irradiance it may select electrically convenient but energetically unrepresentative trackers when the irradiance field is strongly clustered.

2.4.5. Simulated annealing

The present algorithm normalizes all variables and restricts each neighbor move to remain inside the same geometric cluster, accelerating convergence while preserving diversity. Comparable annealers have recently increased PV hosting capacity in LV feeders [25]. Although the method rapidly improves the score at high temperature, its final quality is sensitive to the cooling rate: too aggressive a schedule freezes exploration prematurely, whereas an overly mild one inflates run-time without further benefit.

2.4.6. Multi-objective –Geometrical

This three-objective NSGA-II balances dispersion, slope homogeneity and distance homogeneity. [9] reported that combining these geometric metrics produced Pareto fronts whose extreme points were readily interpretable by PV engineers. Nevertheless, because irradiance is absent from the fitness vector, the analyst still must pick a Pareto member that performs acceptably in energy terms.

2.4.7. Multi-objective –Irradiance

Replacing slope and distance by the relative error against simulated *G_{eff}* links the search directly to energy. Recent facade-PV studies confirm that such bi-objective formulations give layouts that reconcile daylighting and electricity output [24] If the error metric and dispersion are highly correlated, however, the Pareto front may collapse to a narrow band, reducing the designer's freedom.

The hyperparameter configuration of the different algorithms is summarized in Table 1.

By implementing a diverse set of selection algorithms, we aimed to capture the trade-offs between computational simplicity, spatial coverage, and energetic representativeness. Heuristic methods such as K-Means and Maximum Dispersion offer fast, interpretable solutions but may overlook critical irradiance dynamics. In contrast, metaheuristic approaches like Genetic Algorithms and NSGA-II provide more flexible and performance-driven placements at the cost of increased computational effort. The Multiobjective-irradiation method uniquely focuses on maximizing representativeness based on simulated plant performance, providing a perspective aligned with past studies [26]. The effectiveness of each approach is quantitatively evaluated in Section 3.

2.5. Benchmark

The benchmark used in this study is a set of various utility-scale photovoltaic projects located in Spain. Installed capacity of the projects ranges from 30 MW to 70 MW. The sites are chosen because of their

highly mountainous terrain with significant topographic variation. This poses a challenging scenario for sensor placement due to shading effects and uneven irradiance distribution. More details of the projects can be found in Table 2.

Projects A and B are located nearby. They have the same configuration with single-axis tracker, module, module power of 530 W and string inverter. Tracker specifications include $\pm 60^\circ$ rotation range, 2-meter module height at horizontal position, and 0.4 ground coverage ratio.

Project C is located in the region of Málaga, and features a following-terrain tracker (one string per section of tracker). With module power of 530 W and also string inverter.

The main differences is the topography of the sites. The histogram of different slopes of the tables of each project is provided below.

In the North–South slope direction (Slope_NS, upper panel), all three projects display a considerable proportion of tables in intervals above 8 %. Although Projects 1 and 2 show a moderate peak within the [0–4 %] bins—with more than 40 % of their tracker tables in these lower ranges—they still present a substantial percentage of tables in intermediate and high-slope intervals (above 8 %), which is unusual for this type of installation. Project 3 stands out for its even more pronounced topographical variation: over 60 % of its tables are distributed across slopes greater than 8 %, with significant representation in ranges reaching up to 16 % and even 28 %. This highlights a complex landscape with challenging gradients for solar installation and operations.

In the East–West direction (Slope_EO, lower panel), all projects again display non-negligible fractions of tables in intermediate and high-slope intervals, despite the expected flatness in this axis for most utility-scale PV layouts. While the modal bin remains the [0–2 %] range—especially for Projects 1 and 2—there is clear evidence of dispersion toward higher slopes in Project 3, where up to 30 % of the tables are located in ranges above 6 %.

To capture data from actual projects, data is retrieved from SCADA system of the plants. In this case Project B has been used for comparison due to data availability.

The main characteristics of the sensors are detailed below in Table 3 and Fig. 2 (Table 4.).

2.6. Evaluation metrics

Several metrics were defined to evaluate the quality of each method:

- Mean Absolute Error (MAE) quantifies the average absolute deviation between the irradiance measured by the selected pyranometers $G_{selected}$ and the actual mean irradiance across the whole plant G_{plant} . A lower MAE indicates better pointwise accuracy in representing the global irradiance profile.
- $MAE = \frac{1}{T} \sum_{t=1}^T |G_{selected}(t) - G_{plant}(t)|$

Table 1
Hyperparameter configuration

ID	Core algorithm	Main hyper-parameters
1	Lloyd K-Means	$k = N_{sens}$; $n_{init} = 10$; $max_iter = 300$
2	2-stage K-Means	Stage 1: idem K-M; Stage 2: per-cluster score
3	Ward agglomerative (Hierarchical clustering)	Linkage = Ward; Metric = Euclidean
4	Maximum Dispersion	None (deterministic); distance penalized by slope & dist. weights = 1
5	Simulated annealing	$T_0 = 1000$; Cooling = geom. 0.99; Max iters = 1000
6	Multiobjective- geometrical	Pop. = 200; Generations = 100; Crossover = cluster-aware, $pcx = 0.5$; Mutation = within-cluster, $p_{mut} = 0.2$
7	Multiobjective-irradiance	Pop. = 100; Generations = 200; same pcx , p_{mut}

Table 2
Selected Project details.

Code	Installed Capacity (MWp)	Location	Tracker	Electrical configuration	Number of albedometers
Project A	30	Alcalá de los Gazules (Cádiz)	Single Axis NS	String inverter	5
Project B	70	Alcalá de los Gazules (Cádiz)	Single Axis NS	String invertir	5
Project C	50	Málaga	Terrain-Following NS	String inverter	5

Table 3

Data of tables slope.

NS Slope Range (%)	NS Slope (% over total)			EW Slope (% over total)		
	Project 1	Project 2	Project 3	Project 1	Project 2	Project 3
0–2 %	19,49	26,29	12,00	39,91	34,88	8,57
2–4 %	21,32	23,39	12,16	8,92	13,69	8,75
4–6 %	24,91	16,24	10,78	4,10	5,14	9,51
6–8 %	12,61	12,95	10,47	2,43	2,38	9,38
8–10 %	8,27	7,85	10,18	1,70	0,81	7,12
10–12 %	7,18	7,05	12,81	0,47	0,31	4,00
12–14 %	2,82	5,71	11,56	0,11	0,07	2,44
14–16 %	1,93	0,52	9,14	0,07	0,04	1,35
16–18 %	1,48	0,00	5,82	0,01	0,01	0,55
18–20 %	0,00	0,00	2,39	0,00	0,01	0,23
20–22 %	0,00	0,00	1,61	0,00	0,01	0,03
22–24 %	0,00	0,00	0,86	0,03	0,00	0,00
24–26 %	0,00	0,00	0,21	0,03	0,00	0,00
26–28 %	0,00	0,00	0,03	0,01	0,00	0,00
28–30 %	0,00	0,00	0,00	0,01	0,00	0,00

- Mean Relative Error (MRE) provides a normalized percentage measure of the deviation between selected and actual irradiance. This metric is useful when comparing performance across plants of different irradiance magnitudes, as it reflects the proportional accuracy of the estimation.
- $MRE = \frac{1}{T} \sum_{t=1}^T \frac{|G_{selected}(t) - G_{plant}(t)|}{G_{plant}(t)} \times 100\%$
- Coefficient of Determination (R²) assesses how well the temporal pattern of the selected pyranometers explains the overall variability of the plant's irradiance. High R² values indicate strong correlation in dynamic behavior across time, which is essential for performance ratio (PR) analyses.
- $R^2 = 1 - \frac{\sum_{t=1}^T (G_{selected}(t) - G_{plant}(t))^2}{\sum_{t=1}^T (G_{plant}(t) - \bar{G}_{plant})^2}$
- Mean Inter-pyranometer Correlation evaluates the temporal redundancy between selected sensors. A lower mean correlation suggests that the selected pyranometers capture different irradiance dynamics, improving overall representativeness.
- $\rho_{mean} = \frac{1}{N(N-1)/2} \sum_{i < j} corr(G_i(t), G_j(t))$

These metrics provide complementary assessment perspectives: MAE quantifies absolute deviation, MRE normalizes for cross-plant comparison, R² evaluates temporal correlation essential for performance ratio analyses, and inter-sensor correlation assesses redundancy where lower correlation suggests better coverage of different irradiance dynamics.

These metrics were computed over hourly time series of effective irradiance (G_{eff}) for 288 representative hours across the year. Hourly resolution was chosen because it balances temporal resolution and computational cost. Nighttime hours were excluded, and temporal gaps were inserted between days to avoid misinterpretation in graphical representations.

The results enable comparison of both spatial and temporal representativeness of each method, highlighting the strengths and limitations of geometry-based approaches versus those incorporating simulated energetic data.

3. Results

This section presents the outcomes of the proposed selection algorithms, comparing their spatial behavior and representativeness based on the metrics described in Section 2.5. Visual and quantitative analyses are used to evaluate the ability of each method to capture the effective irradiance behavior of the plant.

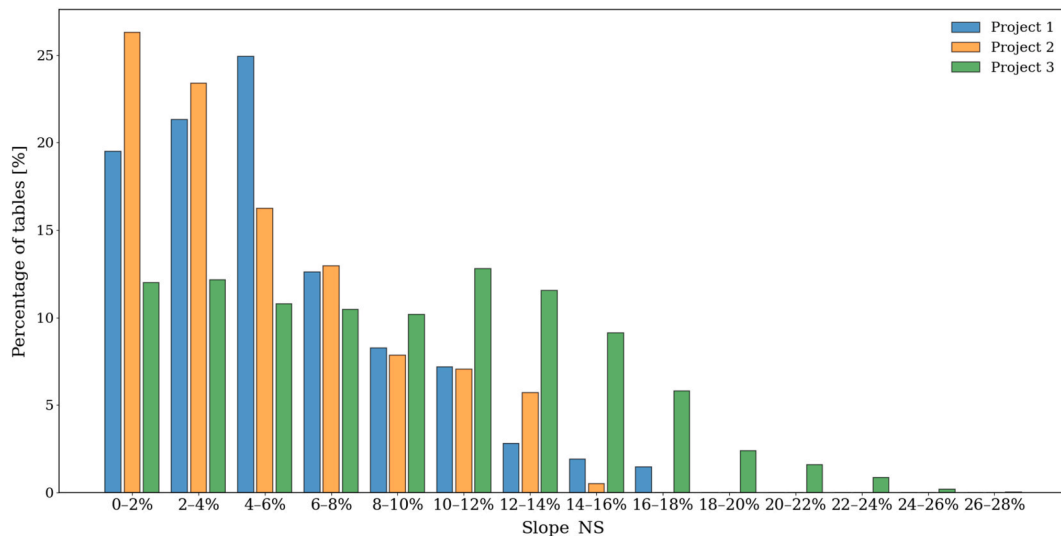
3.1. Irradiance model validation

First, the model has been tested against sensors in the plants. Data from albedometer fixed to the torque tubes have been extracted and compared with simulated irradiation from trackers. Albedometers are usually fixed to torque tubes as depicted in Fig. 2. Since we are using

Table 4

Sensor specifications.

Model	Hukseflux SR30-M2-D1
Standard	IEC 61724-1 compliance meets Class A PV monitoring system requirements
Calibration uncertainty	< 1.2 % (k = 2)
Rated operating temperature	−40 to + 80 °C range

**Fig. 2.** Histogram of North-South slopes.

functions that compute rear irradiation with approximate models that considers shadows from nearby trackers and the albedometer is usually placed in the extreme point of the tracker, it is expected to see a difference in the rear side irradiation.

To ensure a valid comparison between simulation and measured data, it is necessary that the backtracking algorithm applied in the simulation accurately reflects the control strategy implemented by the on-site tracker system connected to the albedometer. However, this alignment is challenging because state-of-the-art single-axis trackers frequently employ proprietary or adaptive backtracking strategies, which can differ substantially from the standard astronomical or 2D geometric tracking algorithms typically used in simulation software [27].

Modern trackers may utilize terrain-adaptive or 3D backtracking approaches that dynamically adjust the tracker angle in response to local row-to-row shading, ground slope, and installation deviations—factors which are usually simplified or neglected in idealized models. As a result, discrepancies in tracker position under real field conditions versus simulated assumptions can lead to diverging patterns of both direct and rear-side irradiance, especially in the early morning and late afternoon or in sites with non-uniform topography. This underlines the importance of both knowing and implementing the specific backtracking logic of the installed tracking system in any quantitative validation analysis.

For the comparison, data for an inclined albedometer from project C and a tracker nearby is used. Data reflects the day of 21st May of 2025, a sunny day with no clouds, replicating the clearsky model used in the simulation. Position of the tracker during the day can be found in Fig. 3.

As can be seen, while in the central hours of the day the position remains close, there are big differences in the early and late hours. While the simulated tracker remains in the last position used, the actual tracker starts to change its position earlier. This behavior can be caused by the backtracking strategy of the tracker or a defensive position caused by wind or another meteorological event. This leads to a difference in the irradiance captured that can be seen in Fig. 4.

As can be seen, there is good agreement in the irradiance on the front side of the panel. Main differences can be found at the extreme of the curve and are more likely due to the difference in backtracking strategy.

Something more interesting happens with *G_{rear}*, that is notoriously asymmetric with respect to the albedometer data.

These discrepancies, reaching up to 50 % in some cases, are attributed to:



Fig. 4. Pyranometer used in Project B.

Measurement location bias: Albedometers are mounted on tracker torque tubes in relatively clear areas, receiving more reflected radiation than modules located in interior rows

pvinfos model limitations: The assumption of planar terrain doesn't account for the complex 3D geometry of mountainous installations and the SVF methodology may not capture the terrain influence properly.

Tracker position differences: Discrepancies in actual vs. simulated tracker positions affect rear-side view factors

Overall, the results show good agreement in *G_{front}*, with a relative error close to 1 % in the central hours of the day but discrepancies in *G_{rear}*, with discrepancies of 50 %. These differences are probably due to the mentioned causes. Since sensor placement is primarily driven by spatial gradients in effective irradiance rather than absolute energy yield, we consider the model valid for the intended application. Nonetheless, future work should investigate improved terrain-aware rear-side modeling to enhance reliability for quantitative energy forecasting.

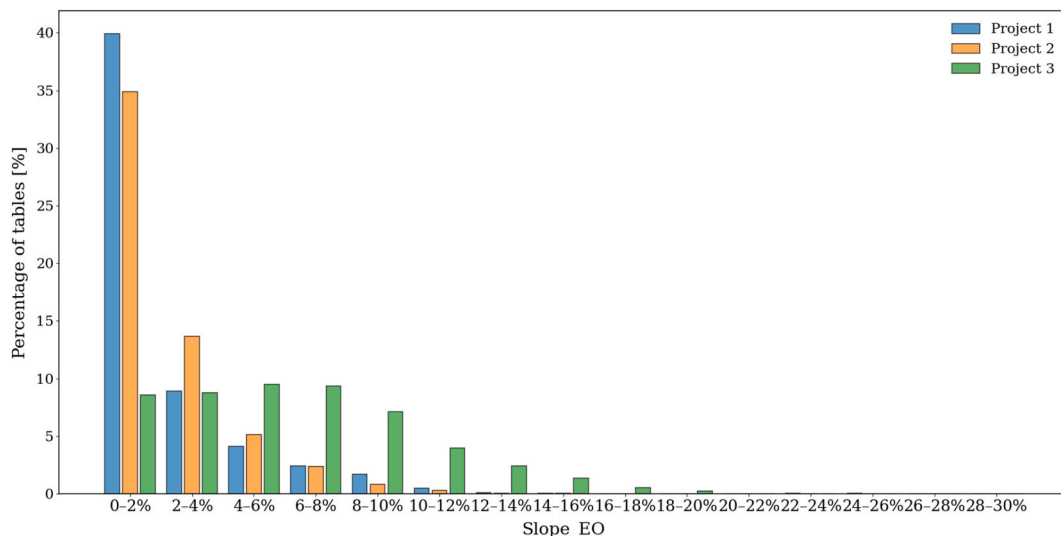


Fig. 3. Histogram of East-West slopes.

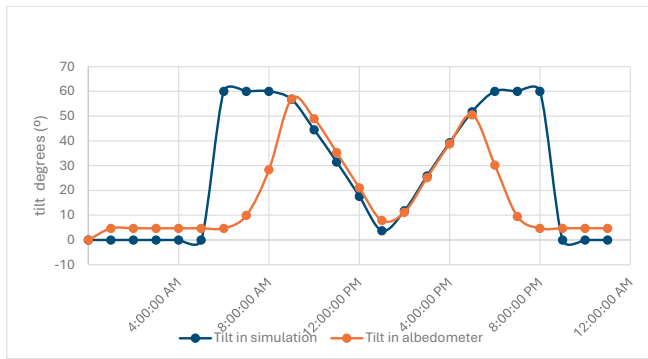


Fig. 5. Comparative of tracker position between model and real tracker for the 21st May 2025.

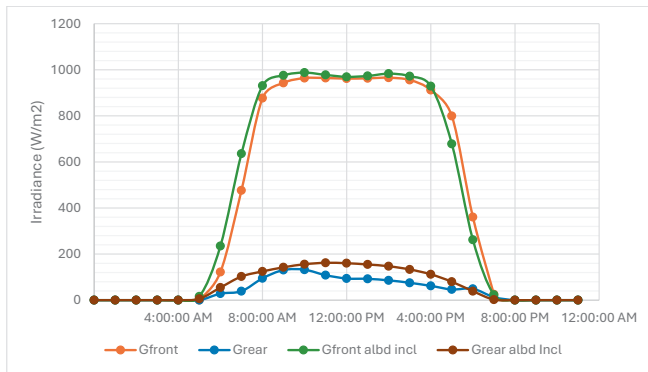


Fig. 6. Comparative of Gfront a Gear of model and POA albedometer for the 21st May 2025.

3.2. Algorithm assessment

Fig. 5 presents the average relative error obtained by each method. Multi-objective methods—both geometric and energetic—achieve the lowest relative errors, indicating better energetic representativeness. Notably, K-Means, Hierarchical clustering and Max Dispersion show

greater relative deviations, supporting the hypothesis that these are suboptimal in complex terrain. Multi-objective algorithms on the other hand tend to represent global irradiance more accurately (Fig. 6, Fig. 7).

Key findings include:

Multi-objective irradiance optimization (MO-irradiance) delivers the most reliable performance across diverse terrains.

In every project the MO-irradiance algorithm maintained mean-relative errors within $\pm 1.9\%$ and mean-absolute errors below 16 W m^{-2} , while preserving temporal correlations of at least 0.98. Its explicit inclusion of plane-of-array error as an objective allows the optimizer to adapt to site-specific shading patterns without sacrificing spatial dispersion, making it the only method that generalizes seamlessly to all three case studies.

Geometry-only algorithms can equal energy-aware methods on individual sites but lack transferability.

Refined K-Means and Ward hierarchical clustering achieved MAE values as low as $5\text{--}9 \text{ W m}^{-2}$ on Project C, highlighting that slopes, distances and coordinates already encode much of the irradiance heterogeneity. However, their errors fluctuated by up to $25,91 \text{ W m}^{-2}$ when applied to Project A, underscoring the sensitivity of purely geometric heuristics to local terrain morphology.

Multi-Objective geometrical algorithm, however, generalizes well across three projects, with similar performance than multi-objective irradiance optimization model. This means that a multi-objective algorithm based on geometry could be as useful as one based on an irradiation method. This is significant, because it is way easier to implement.

The greedy Max-Dispersion heuristic excels at spatial coverage yet incurs systematic energy bias under complex shading.

By maximizing pairwise tracker distance it guarantees excellent geometric spread, but in Project C underestimated effective irradiance by -4.92% and posted the worst MAE ($\approx 16,98 \text{ W m}^{-2}$). This confirms that distance maximization alone cannot compensate for directional shading from surrounding relief.

Simulated Annealing offers a pragmatic speed-accuracy compromise. When its neighborhood is restricted to within-cluster swaps, SA converges in roughly one-tenth the evaluations required by GA/NSGA-II, while preserving $R^2 \geq 0.99$ and correlation ≥ 0.99 in two out of three projects. Although its MAE is slightly higher than MO-

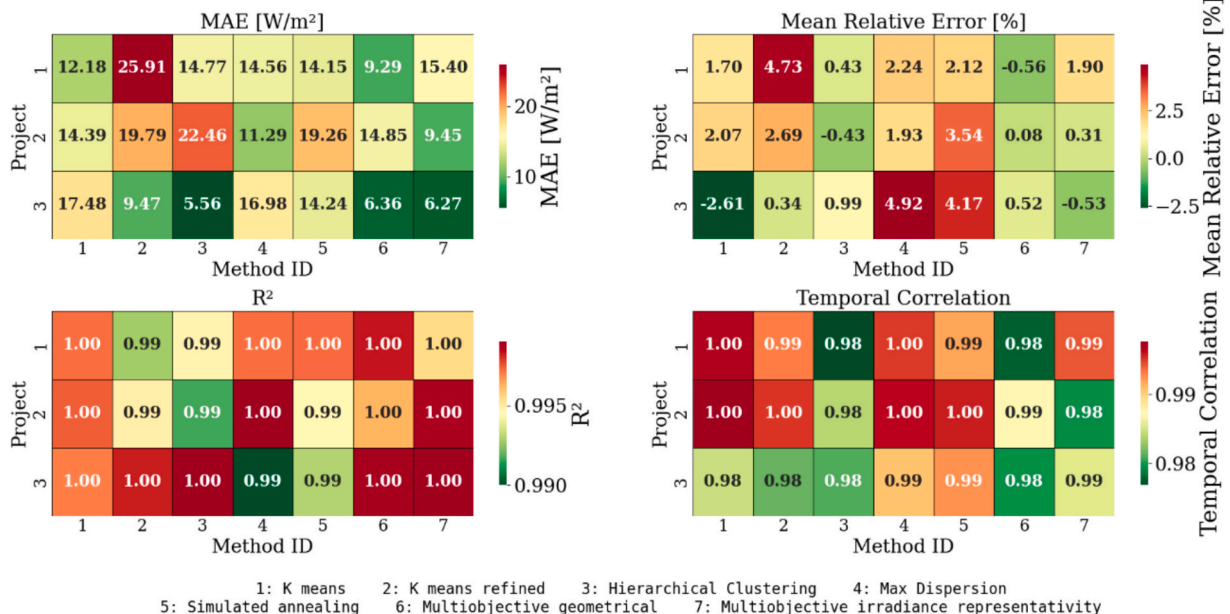


Fig. 7. Main results of the different algorithms (axis X) for the three projects assessed (axis Y). IDs are detailed in Table 1.

irradiance, the reduced run-time makes it attractive for iterative design studies or real-time re-optimization. A hybrid two-stage workflow is recommended for large projects. First, a fast geometric pre-filter—such as Refined K-Means—generates feasible candidate sets that ensure topographic diversity and wiring feasibility. Second, MO-irradiance refines those candidates to minimize energy reconstruction error, achieving site-independent accuracy without prohibitive computational effort. This hybrid strategy capitalizes on the strengths of both geometric and energy-driven perspectives, providing a scalable template for modern PV-plant monitoring design. The generalization capacity of the MO-irradiance and MO-geometry algorithms suggests that well-balanced optimization of topographic diversity and energetic accuracy is key to robust sensor placement. These methods are less sensitive to site-specific artifacts and are therefore better suited for deployment across heterogeneous portfolios.

3.3. Computational performance

Computations of the full script take place as follows using VS Code, Python 3.13.17.

- **Project A:** 3236,15 s
- **Project B:** 7443.15 s
- **Project C:** 4365,28 s

However, with use of precomputation of SVF factors, irradiance values and raster maps, the calculation of the mentioned items takes places only once and the remaining calculations take place in about 20 % of the time of the full computation (Table 5..)

3.4. Sensor placement Visualization

- **Geometric methods (KMeans, Hierarchical Clustering, Max Dispersion)** tend to distribute sensors evenly across the available area. Their selections broadly cover the spatial extent, prioritizing regular, predictable dispersion and avoiding edge concentrations.
- **Multi-objective and energetically informed strategies (MO-Geometric, MO-Irradiance, Simulated Annealing)** show marked sensitivity to site-specific topographic and energetic features. These methods frequently allocate sensors in zones of complex geometry, high spatial variability.
- **Overlap and divergence:** Although some algorithms converge on similar locations, notably in regular grid project layouts, there is substantial divergence in complex sites demonstrating how algorithm choice can significantly affect coverage and potential monitoring bias (Fig. 8.Fig. 9.Fig. 10..).

4. Conclusions

The comparative study across three utility-scale PV projects confirms that the choice of pyranometer-selection algorithm has a measurable impact on how faithfully a limited sensor set reproduces plant-wide irradiance. By evaluating eight techniques—ranging from heuristic

clustering to multi-objective optimization—under identical geometric, topographic and energetic inputs, the work identifies clear patterns in accuracy, robustness and computational cost.

This study provides a comprehensive benchmarking of pyranometer placement algorithms for utility-scale photovoltaic (PV) plants featuring single-axis trackers and bifacial modules deployed on complex terrain. By integrating both geometric and energy-based methodologies and validating them across three operational sites, several key findings are established:

- **Multi-objective optimization methods, especially those incorporating simulated irradiance patterns, demonstrate superior accuracy and robustness** compared to conventional geometric heuristics for sensor placement. This advantage is particularly pronounced on sites with significant terrain heterogeneity, where irradiance gradients and shadow patterns are highly variable.
- **Geometry-based multi-objective schemes can achieve comparable accuracy to energy-informed models.** This finding is especially relevant for industry practitioners who seek to optimize sensor layouts efficiently, without the need for intensive simulation resources.
- **Validation with on-site SCADA data and albedometers confirms the methodological soundness of the approach** despite known limitations in the modeling of rear-side irradiance: specifically, the use of view-factor methods on non-planar terrain introduces unavoidable uncertainty that should be acknowledged in any quantitative assessment.

5. Practical implications

- The results advocate for the adoption of two-stage optimization strategies in large-scale PV projects, combining rapid geometric pre-filtering with energy-based refinement.
- Design and engineering teams are encouraged to integrate both terrain characterization and detailed sensor placement benchmarking as routine practice in the site assessment and monitoring design process.

6. Future research directions include

- Integration of machine learning techniques for adaptive sensor placement based on operational data
- Extension to agrivoltaic systems where additional constraints (crop access, shading requirements) affect optimal sensor positioning
- Development of dynamic sensor placement strategies that adapt to seasonal irradiance patterns
- Investigation of hybrid monitoring approaches combining permanent sensors with mobile measurement campaigns

7. Data and Code Availability

The code used to implement irradiance modeling and sensor selection algorithms is part of an internal development tool and cannot be made publicly available due to intellectual property restrictions.

However, the methodology is described in sufficient detail to allow independent reproduction of the simulations and selection logic. Key equations, input variables, and algorithm configurations are fully documented in the manuscript and supplementary materials. Synthetic data or simplified examples can be provided upon reasonable request to the corresponding author.

Statement: During the preparation of this work the author(s) used Chat GPT in order to improve readability of the manuscript and certain code functions. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication

Table 5
Computational performance and requirements.

Project	Time of computation (seconds)	Required memory for storing output files (GB)
A	3236.15	4.64
B	7443.15	4.64
C	4365.28	0.8

*For projects A and B the information of the topographic survey was available just merged in one single file.

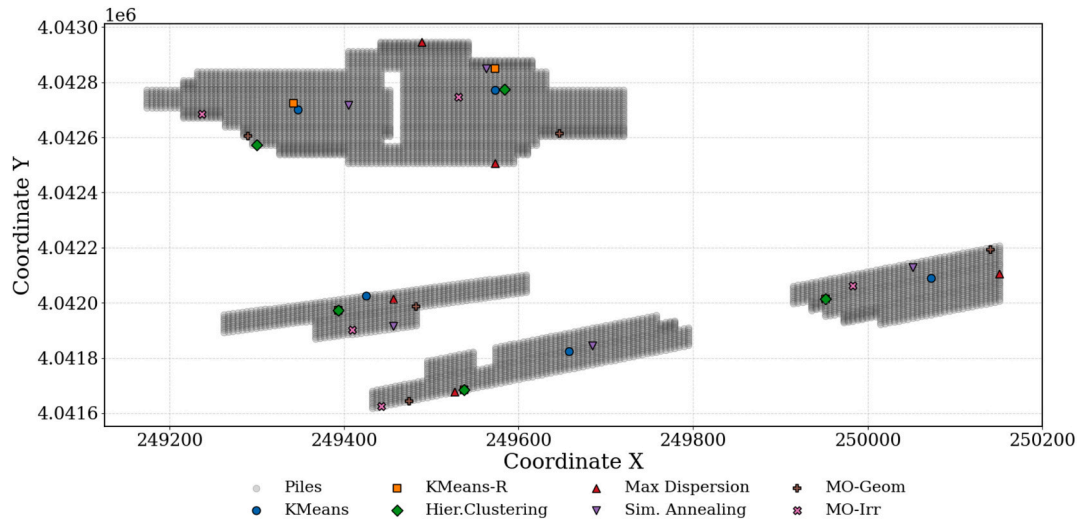


Fig. 8. Project A spatial distribution of sensors.

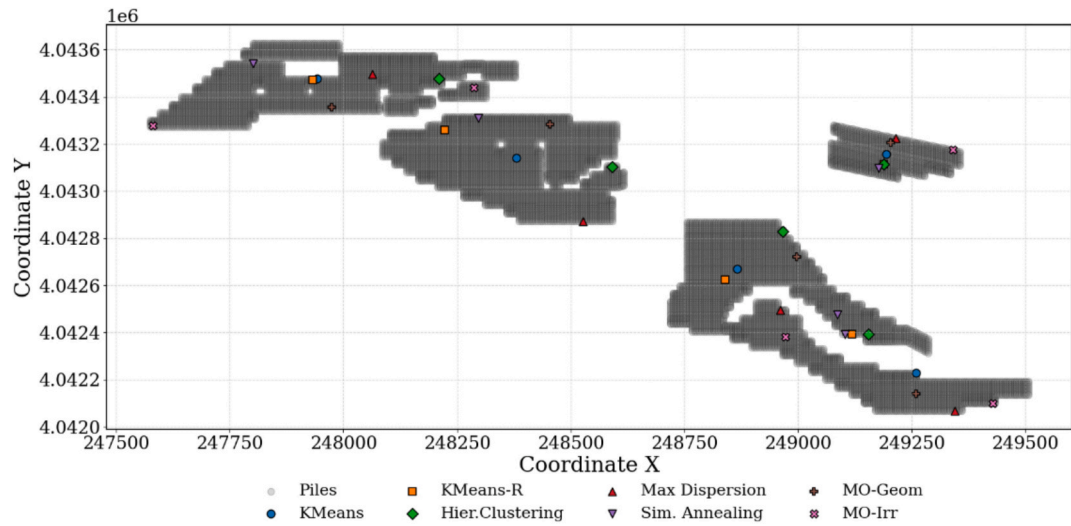


Fig. 9. Project B spatial distribution of sensors.

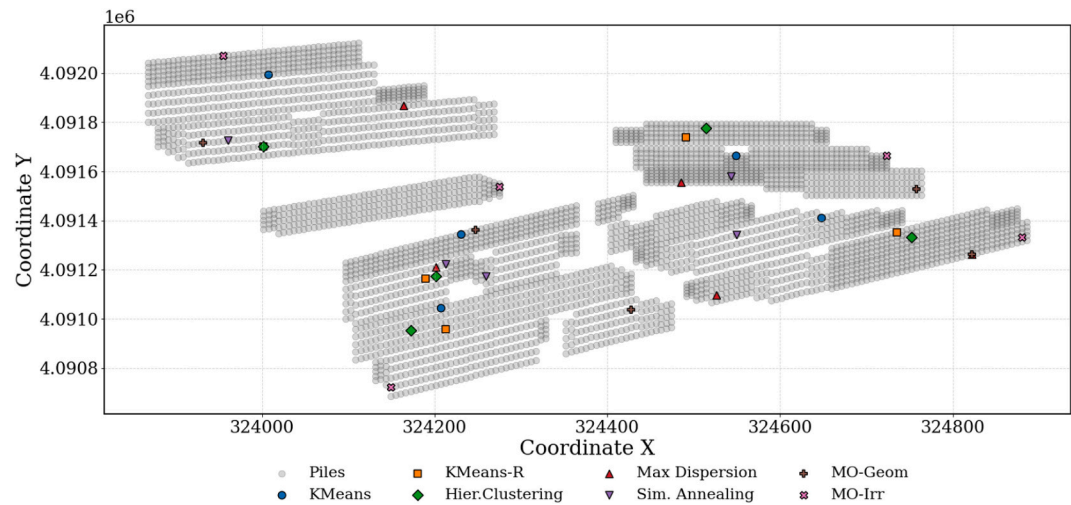


Fig. 10. Project C spatial distribution of sensors.

8. Glossary table

Symbol	Description	Units
<i>G_{front}</i>	Front-side plane-of-array irradiance	W·m ⁻²
<i>G_{rear}</i>	Rear-side irradiance (bifacial contribution)	W·m ⁻²
<i>G_{eff}</i>	Total effective irradiance	W·m ⁻²
SVF	Sky View Factor	–
MAE	Mean Absolute Error	W·m ⁻²
MRE	Mean Relative Error	%
R ²	Coefficient of determination	–
SA	Simulated Annealing	–
MO-irradiance	Multi-objective optimization (irradiance-based)	–
MO-geometrical	Multi-objective optimization (geometry-based)	–

CRedit authorship contribution statement

Conrado García Salinas: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Angel Martín Furones:** Writing – review & editing, Supervision, Conceptualization. **Ana Belén Anquela Julián:** Writing – review & editing, Supervision, Methodology. **Jorge Aleix Moreno:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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